

Artificial Intelligence

Lesson 13: Reinforcement Learning III

Q-Learning

			+10
	WALL	?	-10
●	?		

Transition probabilities fade into question marks.

When the model is unknown, experience becomes the teacher.



Today's mission

Policy and value give meaning to behavior over time.

1

Explain Q-learning and Q-values.

2

Compare model-based and model-free learning.

3

Interpret learning from experience.

Performance target: calculate Q-values with model-free learning and explain how one sampled transition changes one action-value estimate.

Retrieval: what did VI and PI know?

Both previous algorithms planned from a known MDP model.

			+10
	WALL		-10
START			

Known transition model
 $P(s' | s, a)$

Known rewards
 R

Value iteration and policy iteration used these ingredients to compute values and policies.

Then remove the model

What if the agent does not know the transition probabilities?

			+10
	WALL	?	-10
●	?		

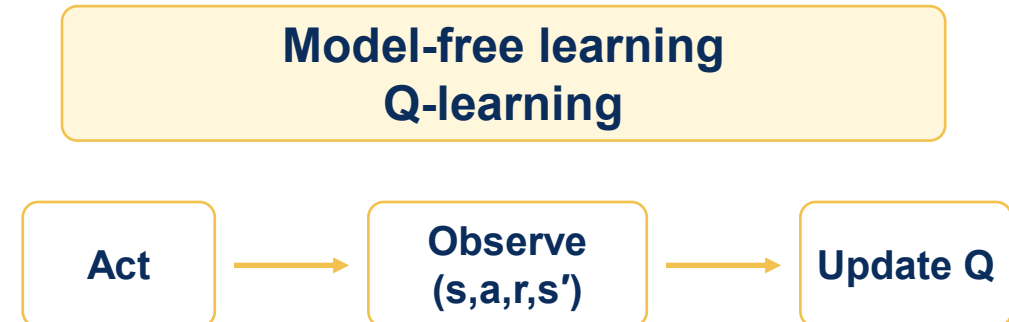
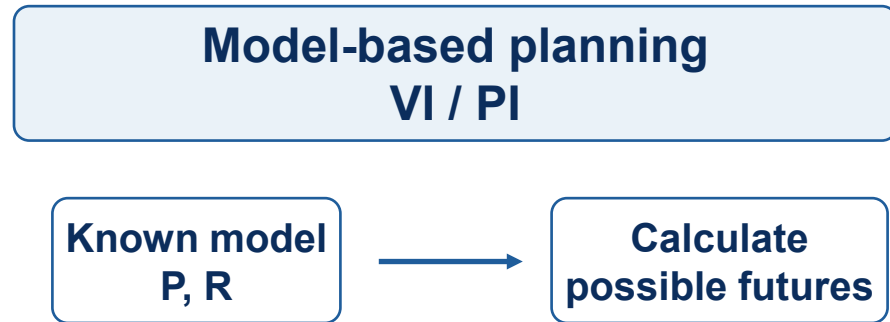
Can we compute
 $\sum P(s'|s, a)V(s')$?

No.
The weights are
unknown.

But the agent can still act, receive rewards, and observe next states.

Planning versus learning

Same decision problem. Different source of information.



Planning reasons over possible outcomes. Learning updates from actual outcomes.

Unknown model $\neq$ no MDP

The world can still be structured even if the agent has not learned the structure yet.

Still exists in the environment

- states
- actions
- rewards
- actual outcomes
- long-term consequences

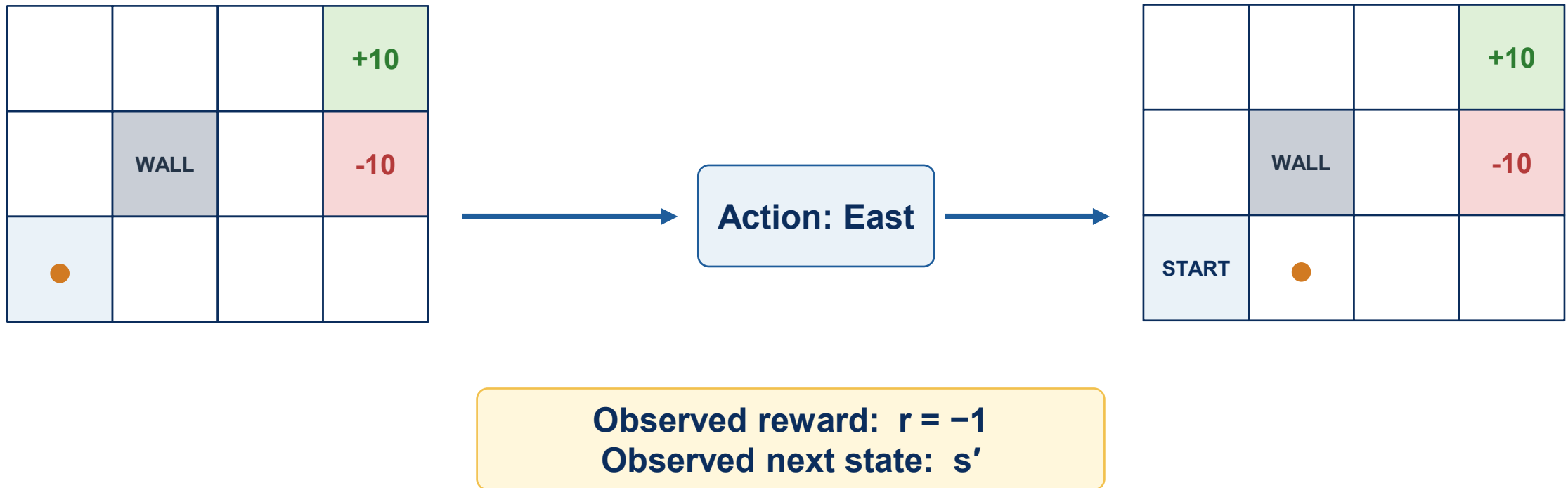
Unknown to the learner

- full transition table
- probabilities for each outcome
- sometimes the reward model
- best actions at the start

Q-learning works by interacting with the MDP instead of requiring the full model in advance.

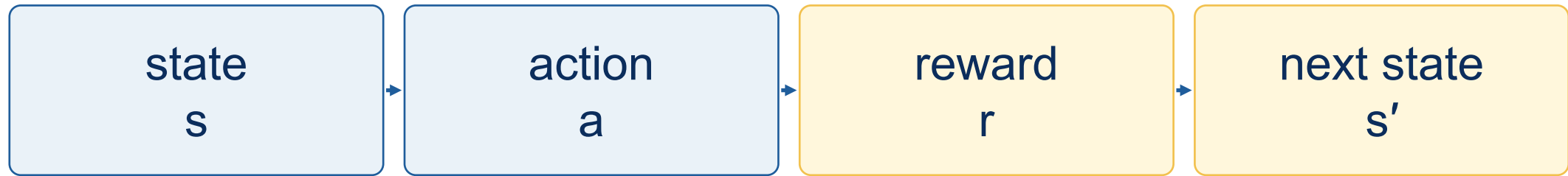
So try the action

One real transition gives evidence.



One interaction creates one experience

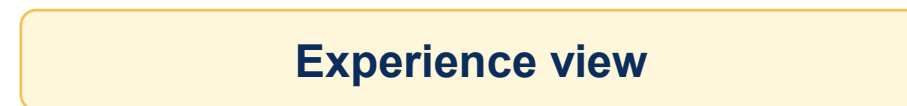
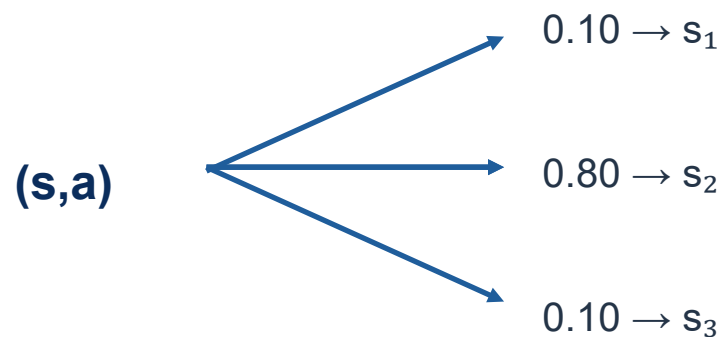
Q-learning updates from the packet (s, a, r, s') .



The learner does not see every possible result. It sees what happened this time.

A sample is not the whole transition model

This is the key model-free shift.



A sample tells us one outcome, not the whole probability distribution.

Why samples can still teach us

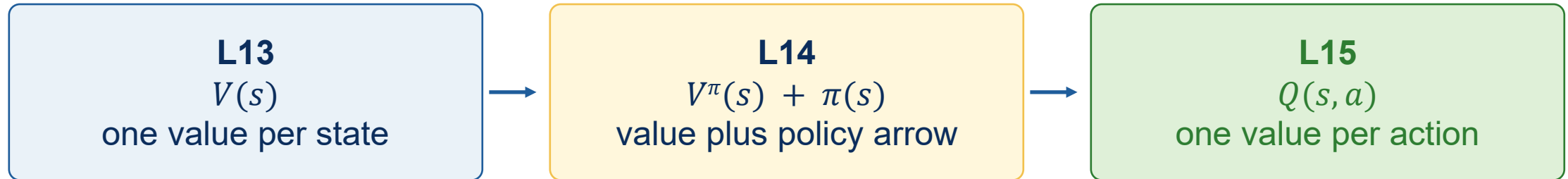
Over time, experience starts to reflect the environment.



**Q-learning does not average over known probabilities;
it lets repeated sampled outcomes influence estimates.**

From state values to action values

The representation changes because the information source changes.



Q-values make action selection possible without explicitly storing transition probabilities.

A Q-value answers: “How good is this action here?”

It estimates long-term return for a state-action pair.



$Q(s, a)$ estimates the long-term return associated with taking action a in state s .

- not a probability
- not necessarily the immediate reward
- not the value of the state alone

A Q-table stores action-value estimates

A greedy action chooses the largest current Q-value.

State	↑	↓	←	→
High battery	4.2	2.5	3.1	5.0
Medium battery	5.8	4.0	3.1	7.2
Low battery	3.6	2.1	1.0	4.4

$$\text{Greedy action} = \arg \max_a Q(s, a)$$

In “Medium battery,” exploitation chooses → because 7.2 is currently largest.

But where do the Q-values come from?

Start with estimates. Let experience change them.

			+10
	WALL		-10
START			

Classroom initialization:

$$Q_0(s, a) = 0$$

Zero means “initial estimate,”
not “the action is truly worthless.”

What can change the estimate?
Experience.

Q-learning starts with an old belief

Before the new experience, we already have an estimate.

Current estimate

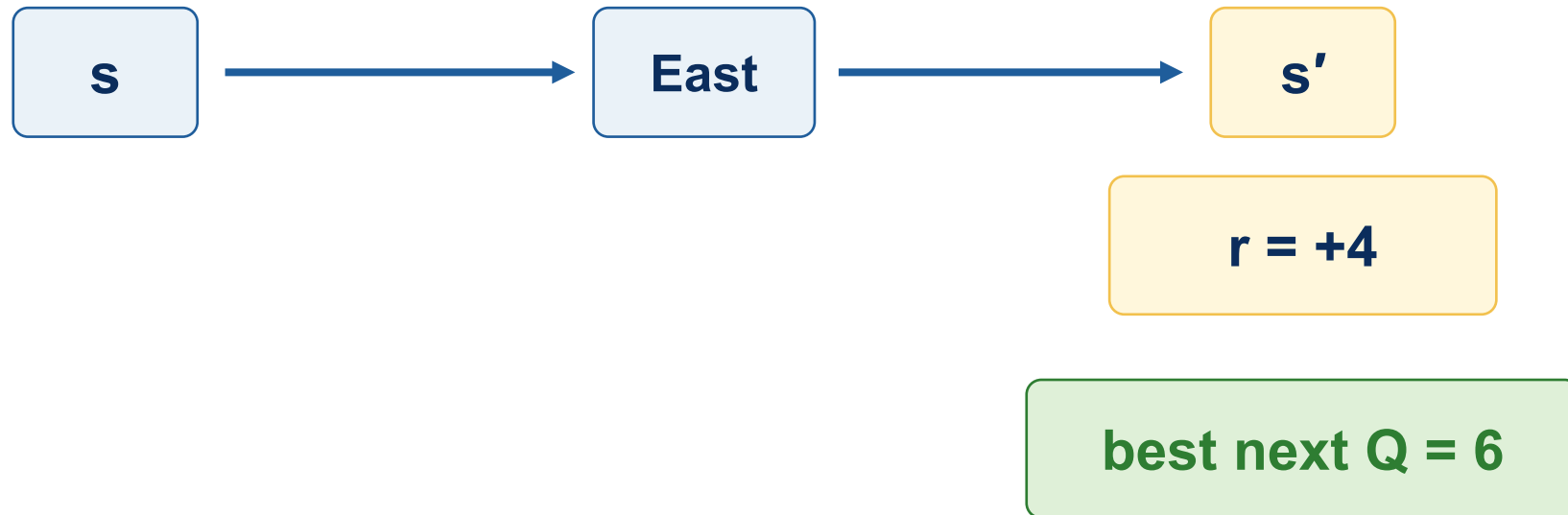
$$Q(s, \rightarrow) = 3$$

Interpretation:
choosing East here is currently estimated
to be worth about 3 in long-term return.



The new experience provides evidence

Observed reward is only part of the target.



Receiving +4 does not mean the long-term value should be 4. Future value still matters.

Build the target

The target is what this experience suggests we should move toward.

$$\begin{array}{|c|} \hline \text{Observed now} \\ \hline r \\ \hline \end{array} + \begin{array}{|c|} \hline \text{Discounted best future} \\ \hline \gamma \max_{a'} Q(s', a') \\ \hline \end{array} = \begin{array}{|c|} \hline \text{Target} \\ \hline r + \gamma \max_{a'} Q(s', a') \\ \hline \end{array}$$

Target = immediate evidence + estimated value of the best next action.

Why MAX but no probability-weighted sum?

Q-learning uses the next state that actually occurred.

Value iteration

$$R + \gamma \sum P(s'|s, a) V(s')$$

All possible outcomes
weighted by known probabilities

Q-learning target

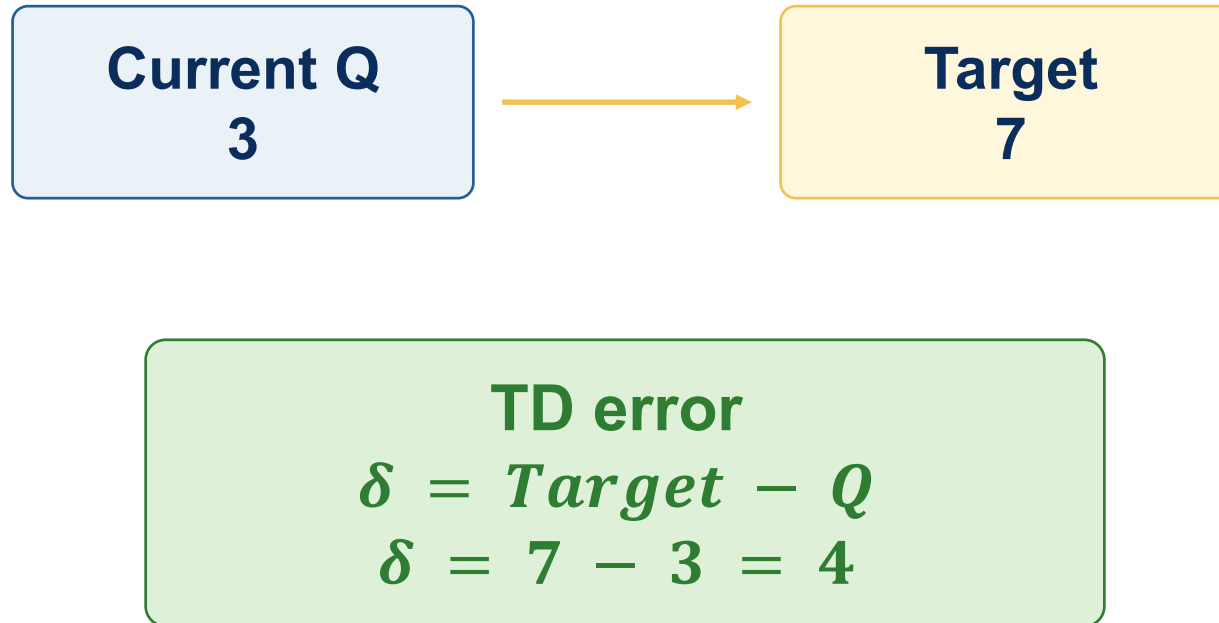
$$r + \gamma \max_{a'} Q(s', a')$$

One next state occurred;
choose the best action from there

No transition probabilities appear in the Q-learning update.

New evidence versus old belief

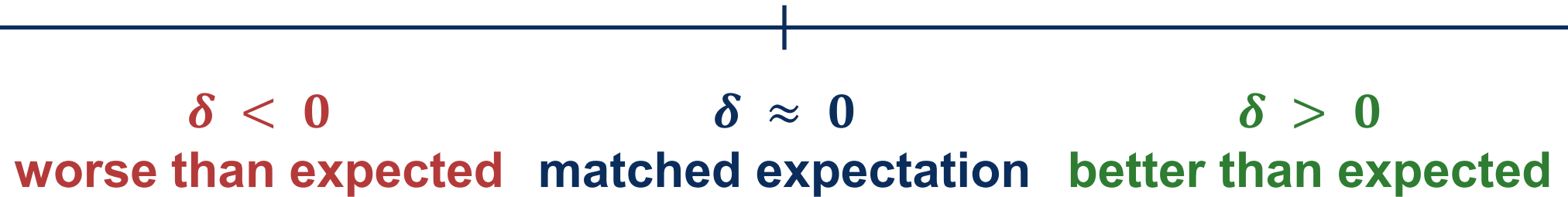
Temporal-difference error measures prediction discrepancy.



The experience was better than our current estimate predicted.

Interpret the sign of TD error

“Error” means discrepancy, not necessarily a programming mistake.



The sign tells whether the target is below, near, or above the current estimate.

Should one experience erase everything?

The learning rate decides how much the sample matters.



Should one sample immediately replace 3 with 7?

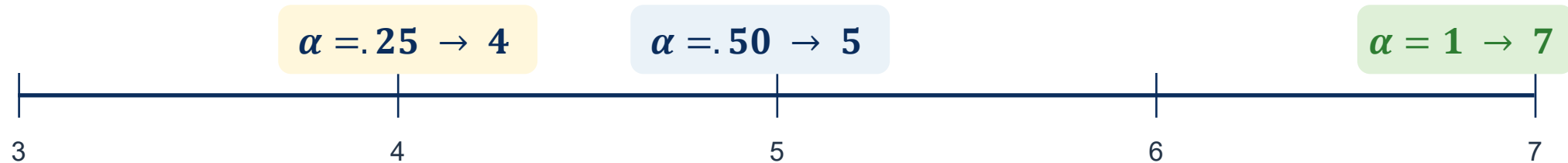
Usually: move partway.

α controls how far we move

Learning rate controls update size.

Current $Q = 3$

Target = 7



Large α : new experiences have more influence. Small α : estimates change cautiously.

Very small α means slow learning

Small is not wrong, but each experience matters less.

Target = 7

$\alpha = .50$
 $3 \rightarrow 5$

$\alpha = .10$
 $3 \rightarrow 3.4$

$\alpha = .01$
 $3 \rightarrow 3.04$

If α is extremely small, learning may require many more samples.

Not every number from 0 to 1 is a probability

Module 2 uses several bounded parameters.

Symbol	Meaning	Probability?
$P(s' s, a)$	transition likelihood	Yes
γ	discount factor	No
α	learning rate	No
ε	exploration probability	Yes

Same numeric range. Different roles.

Put the pieces together

The full Q-learning update is compact notation for a simple adjustment.

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

1 Current belief
 $Q(s, a)$

2 Target
 $r + \gamma \max_{a'} Q(s', a')$

3 TD error
Target – Q

4 Adjustment
 $\alpha \times \text{error}$

The new estimate moves the old estimate toward the target.

Worked update: one experience

Use Compute → Interpret → Predict.

Current Q
 $Q(s,a)=3$

Reward
 $r=4$

Best next Q
 $\max Q(s',\cdot)=6$

Discount
 $\gamma=.5$

Learning rate
 $\alpha=.25$

Task 1: target = $r + \gamma \max Q(s',\cdot)$

Task 2: $\delta = \text{target} - \text{current } Q$

Task 3: $Q_{\text{new}} = Q + \alpha \delta$

Worked update: calculate the pieces

Separate target, TD error, and updated Q.

Target

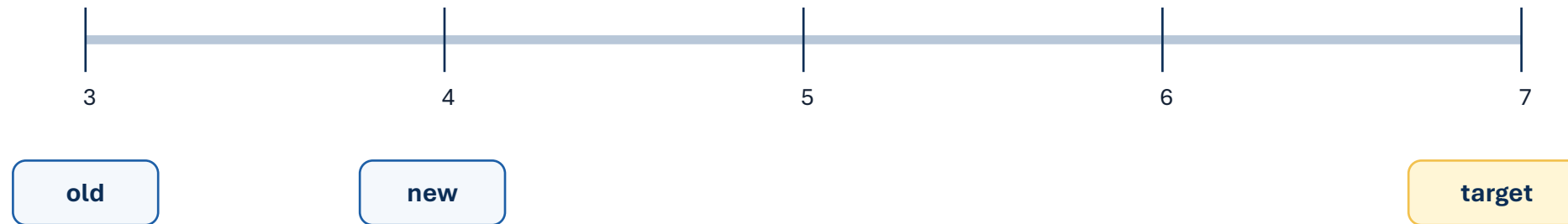
$$4 + .5(6) = 7$$

TD error

$$7 - 3 = 4$$

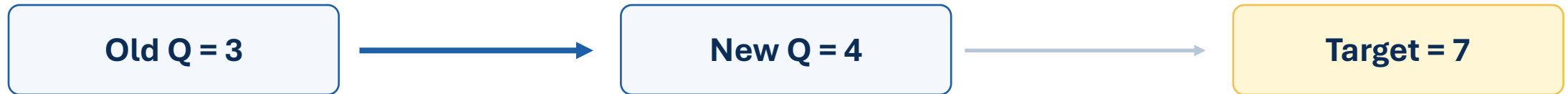
Updated Q

$$3 + .25(4) = 4$$



Interpret the update

The result is a revised estimate, not the immediate reward.



- 1 Why did Q increase?**
The target was larger than the current estimate.
- 2 Why did it not jump to 7?**
The learning rate $\alpha=.25$ moved only partway toward the target.
- 3 What would happen with larger α ?**
The update would move farther toward the target.

Worked update: two experiences

The second update starts from the revised Q-value, not the original estimate.

Same state-action entry: $Q(A, \text{East})$ start $Q = 3.0$ $\gamma = 0.5$ $\alpha = 0.25$

Observed $r=4, \max Q(s', \cdot)=6$

Target $4 + .5(6) = 7$

TD error $7 - 3 = +4$

Update $3 + .25(4) = 4.0$

Interpret Q moves upward

Observed $r=0, \max Q(s'', \cdot)=4$

Target $0 + .5(4) = 2$

TD error $2 - 4.0 = -2$

Update $4.0 + .25(-2) = 3.5$

Interpret Q moves downward

old belief

3.0

after exp. 1

4.0

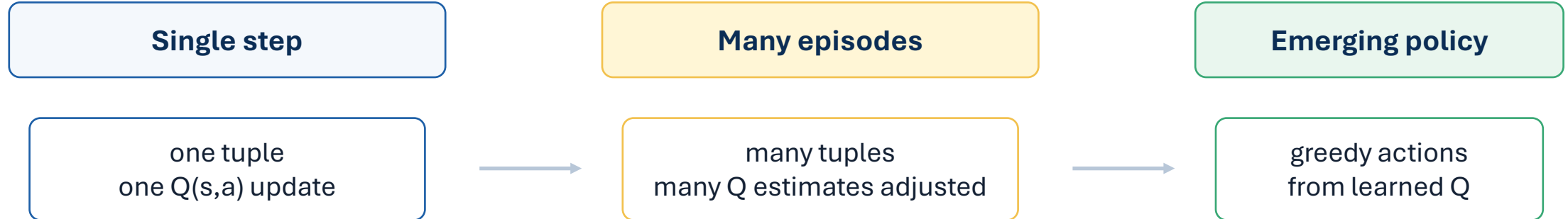
after exp. 2

3.5

Q-learning keeps revising the same estimate as new experiences arrive.

One update is not learning mastery

Q-learning improves through many experiences.



Individual updates can move up or down because sampled outcomes vary.

Q-values across the Gridworld

Experience gradually differentiates state-action estimates.

Episode 0

0	0	0	+10
0	WALL	0	-10
0	0	0	0

After a few episodes

0	0.8	4.5	+10
-0.5	WALL	1.2	-10
0	0.3	0.8	0

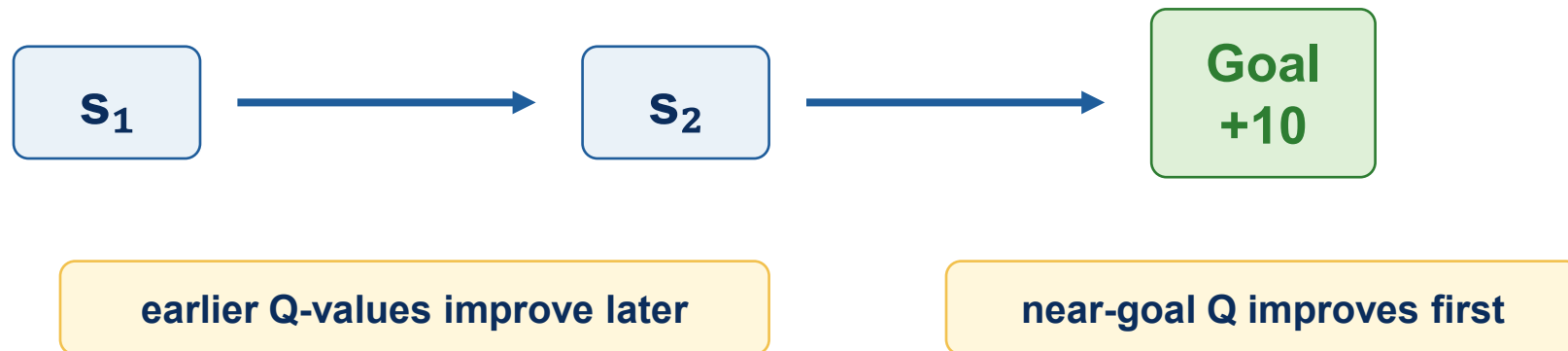
After many episodes

→	→	→	+10
↑	WALL	↑	-10
↑	→	↑	←

A clearer greedy policy emerges as estimates become more informative.

Experience can propagate useful information backward

Repeated transitions carry value through the world.



Value iteration propagated through model-based backups. Q-learning propagates through experienced transitions.

A problem with always choosing the current best

Early estimates can be wrong.

North	8
East	3
South	2

If the learner always chooses North, how will it discover whether East was underestimated?

Exploitation

Use what the learner currently believes is best.

Exploit = choose $\operatorname{argmax}_a Q(s,a)$

Strength

uses what has been learned

Risk

early estimates may be wrong

Exploitation is good for performance, but not always good for gathering information.

Exploration

Sometimes try another action to gather information.

Explore = deliberately sample alternatives

Benefit

may discover better actions

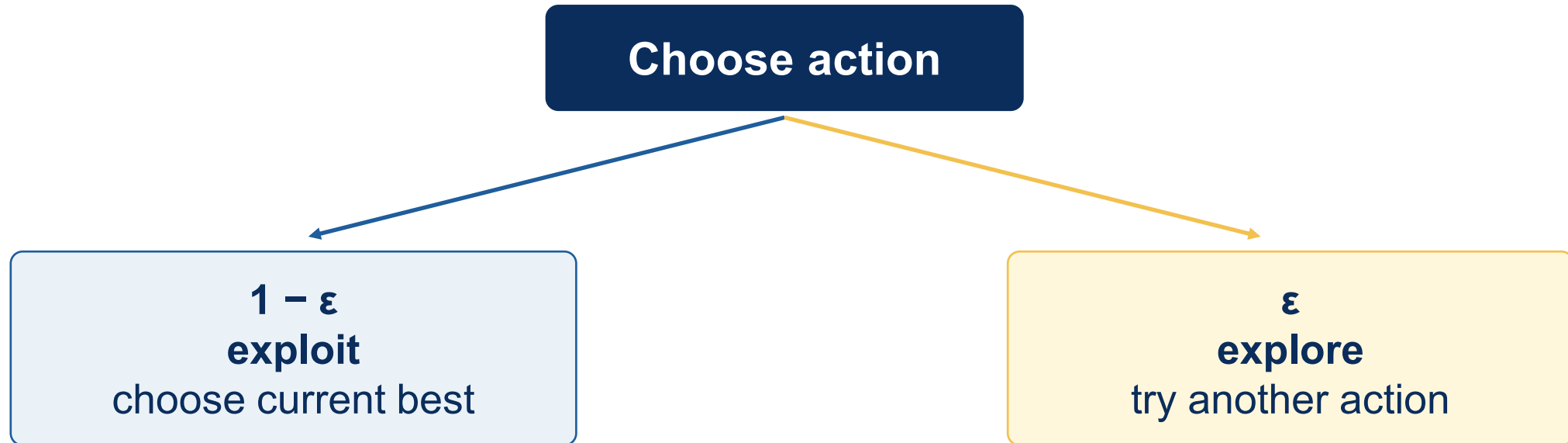
Cost

may temporarily choose worse actions

Exploration gives the learner experiences it would not get by exploiting only.

ϵ -greedy intuition

Mostly exploit; sometimes explore.



ϵ is a probability because it controls how often the learner explores.

Too little vs. too much exploration

Exploration is a tradeoff, not a slogan.

$\epsilon = 0$
never explore
can get stuck

moderate ϵ
mostly exploit
sometimes learn more

very high ϵ
explore constantly
lower performance

Useful learning usually requires both good information gathering and eventual use of what was learned.

More episodes give more opportunities to learn

But experience may still be uneven.



More episodes do not guarantee every state-action pair is sampled equally.

Same Gridworld, three methods

The difference is what information the method uses.

Method	Known model?	Learns from samples?	Main object
Value Iteration	yes	no	$V(s)$
Policy Iteration	yes	no	π and $V^\pi(s)$
Q-Learning	no	yes	$Q(s,a)$

VI and PI plan from a model. Q-learning learns from interaction.

What Q-learning does not require

The update avoids the full transition table.

Crossed out for Q-learning

- × full transition table
- × known $P(s'|s,a)$
- × every possible outcome before learning

Still needed each step

- ✓ observed state s
- ✓ chosen action a
- ✓ observed reward r
- ✓ observed next state s'

Model-free does not mean information-free.

Practice: Q-learning update sprint

Show the target, TD error, and updated Q.

$$Q(s,a)=5 \quad r=1 \quad \max Q(s',a')=7 \quad \gamma=.6 \quad \alpha=.5$$

1. Target = ?

2. TD error = ?

3. Q_{new} = ?

Expected: target = 5.2, $\delta = 0.2$, $Q_{\text{new}} = 5.1$

Is the update upward or downward? What would a larger α change?

Practice: interpret before calculating

Concept first. Arithmetic second.



- Will Q increase or decrease?
- What sign is the TD error?
- Would larger α make the change larger or smaller?

A good learner can often predict the direction before computing the exact value.

Diagnose the mistake

Which claims are correct?

Q-learning needs $P(s'|s, a)$.

Incorrect

The target is just the immediate reward.

Incorrect

Positive TD error means better than predicted.

Correct

Exploration chooses the current highest-Q action.

Incorrect

Larger α gives each sample more influence.

Correct

Read Q-learning

Map each line to the experience-and-update story.

```
initialize Q

repeat:
  observe s
  choose a
  take action
  observe r, s'

  target = r +  $\gamma$  max Q(s', ·)
  error = target - Q(s,a)
  Q(s,a) = Q(s,a) +  $\alpha$ (error)
```

- Where does experience enter?
- Where are transition probabilities?
- Which Q-value is updated?
- Where might exploration influence the algorithm?

Only one experienced state-action pair is updated at a time.

Exit ticket

Complete before you leave.

- Q-learning is model-free because _____.
- The target combines _____ and _____.
- TD error compares _____ with _____.
- α controls _____.
- Exploration is useful because _____.



Next: HW4 due, quiz 4 administered, block 2 workshop/review